

# CrabTransformer Training Report

Scenario: REAL | Loss: MSE | Lag: 0 Years | Current Spawner: No

## — Training Configuration —

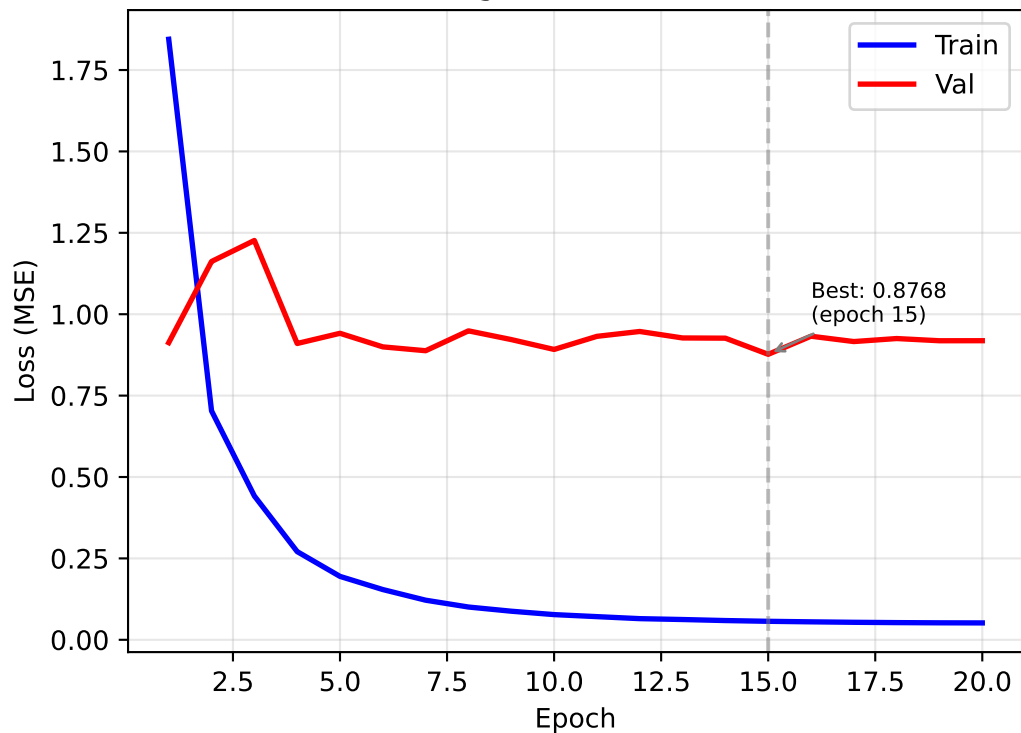
epochs\_run: 20      best\_val\_epoch: 15  
best\_val\_loss: 0.8768      final\_train\_loss: 0.0519  
bias\_correction (MSE): 1.0884  
include\_current\_spawner: False

## — Model Architecture —

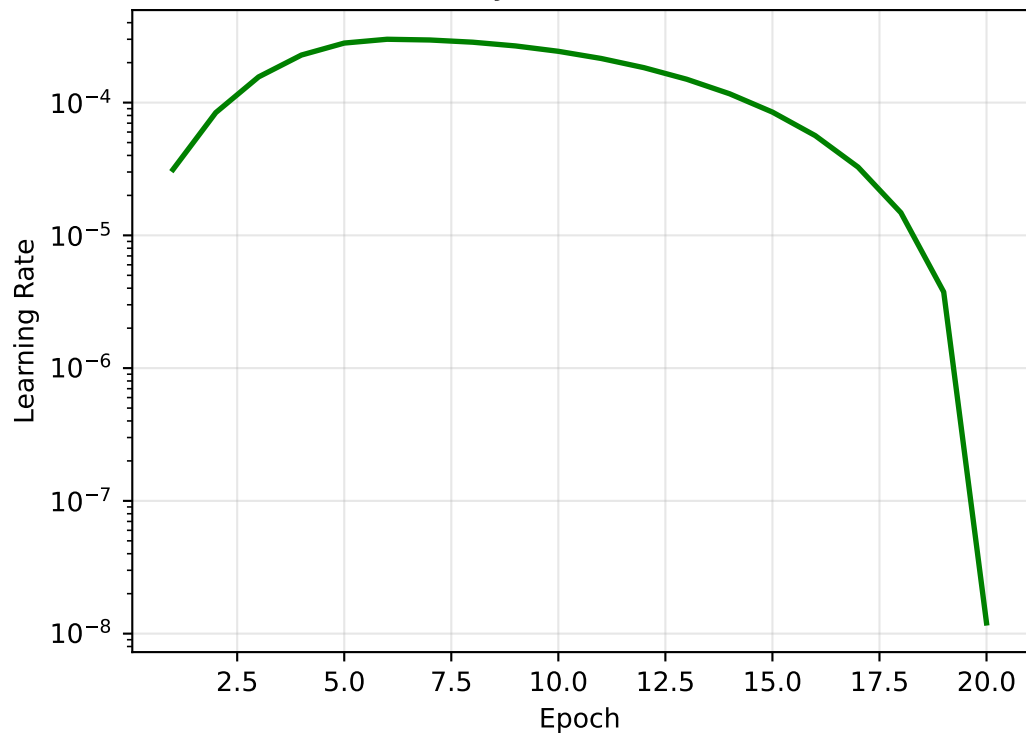
embed\_dim: 128      num\_heads: 8      num\_layers: 6  
d\_ff: 512      dropout: 0.2      patch\_size: 5  
train/val/test years: 24/8/4

# Training Curves: REAL | Lag 0 | MSE | NO Spawner(t)

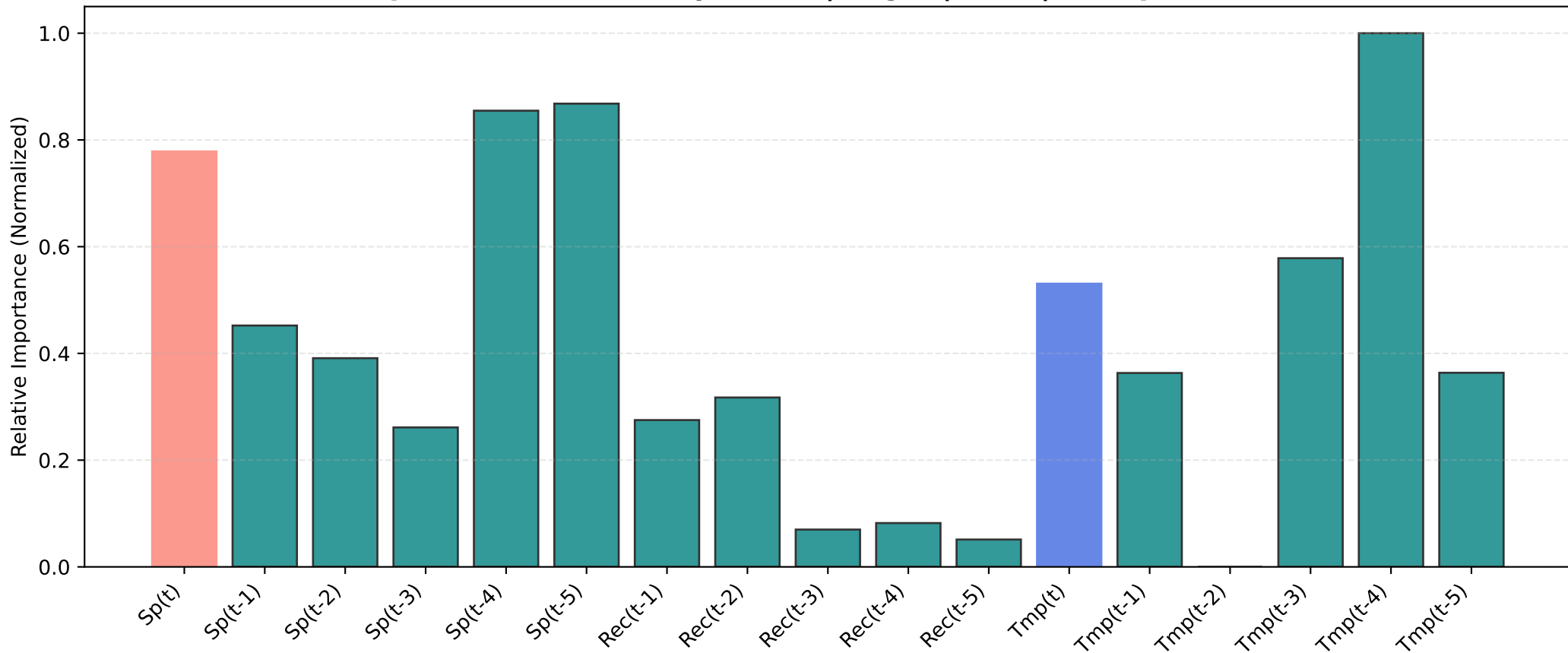
## Training & Validation Loss



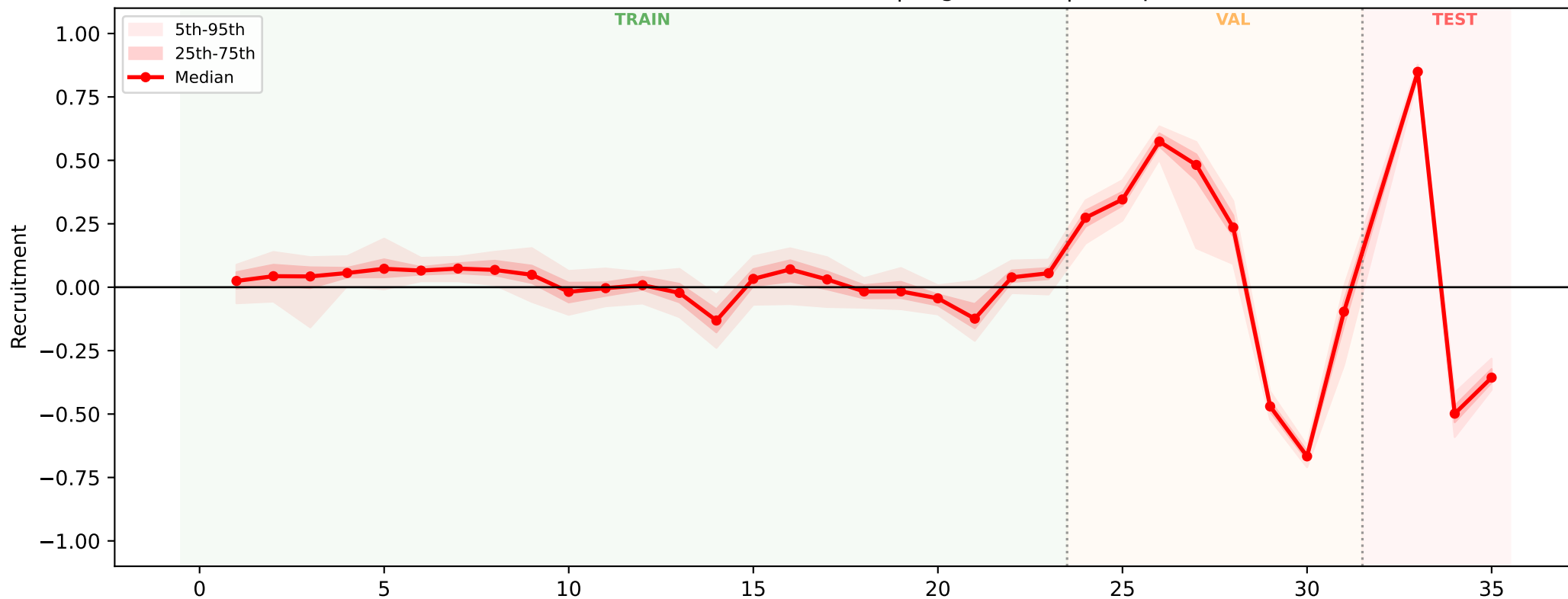
## OneCycleLR Schedule



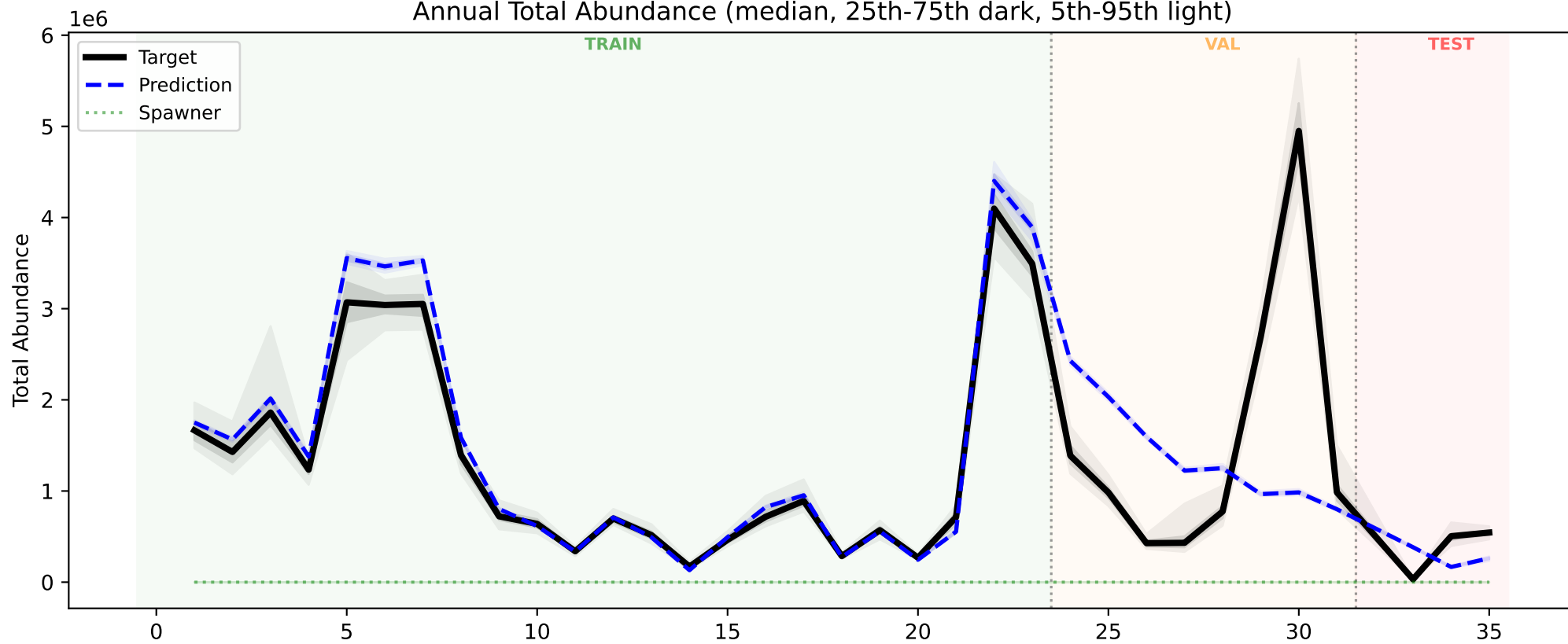
**Input Channel Priority: REAL | Lag 0 | MSE | NO Spawner(t)**



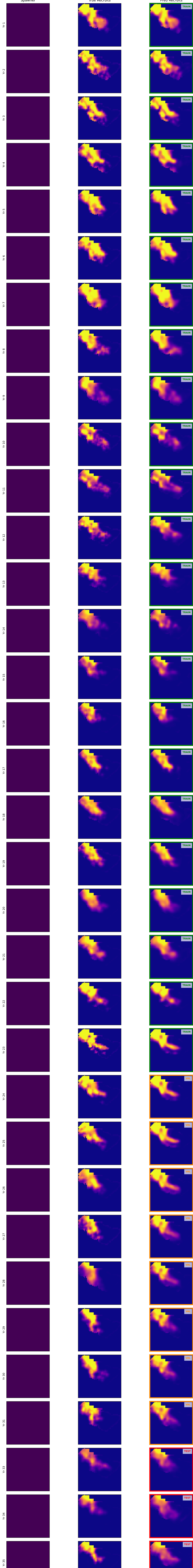
Annual Recruitment Deviations: REAL | Lag 0 (MSE) | NO Spawner(t)



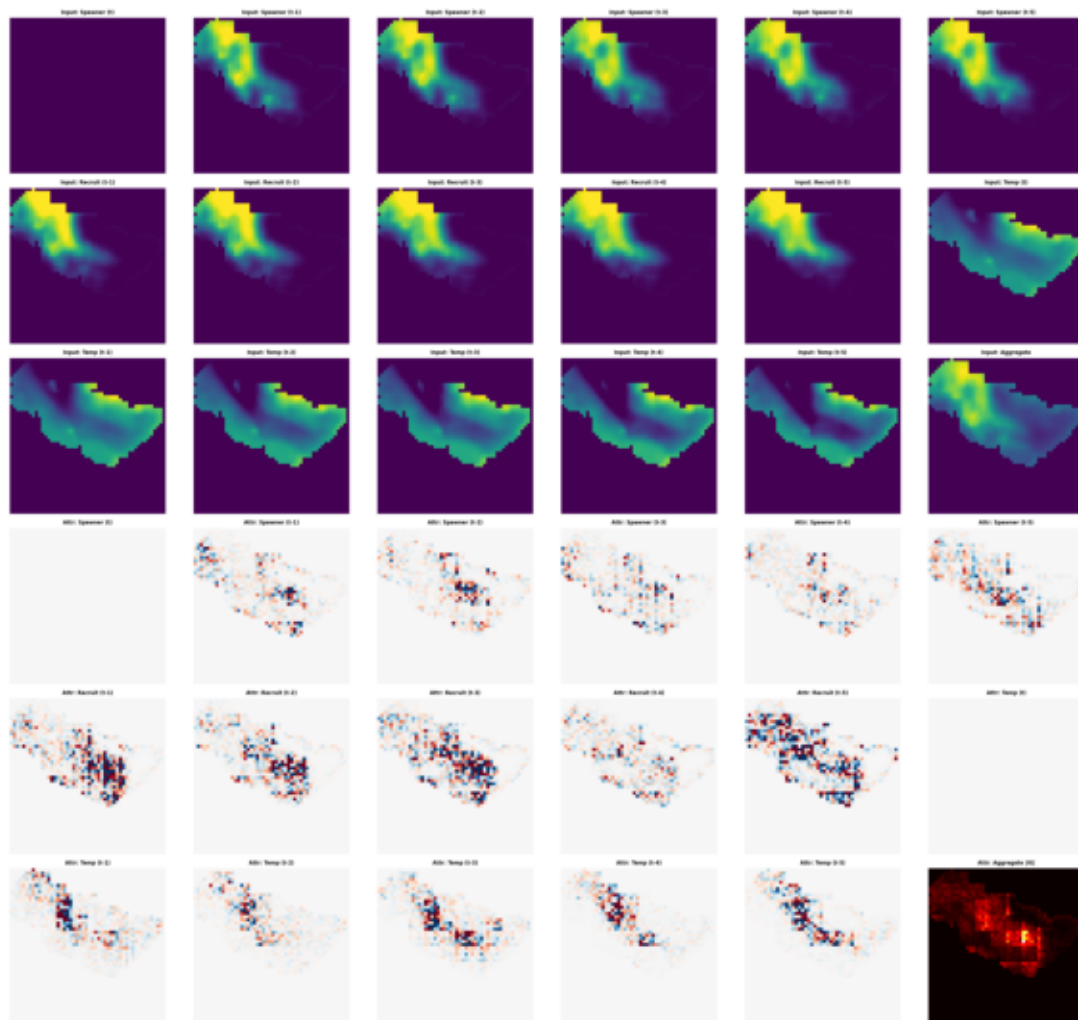
Annual Total Abundance (median, 25th-75th dark, 5th-95th light)



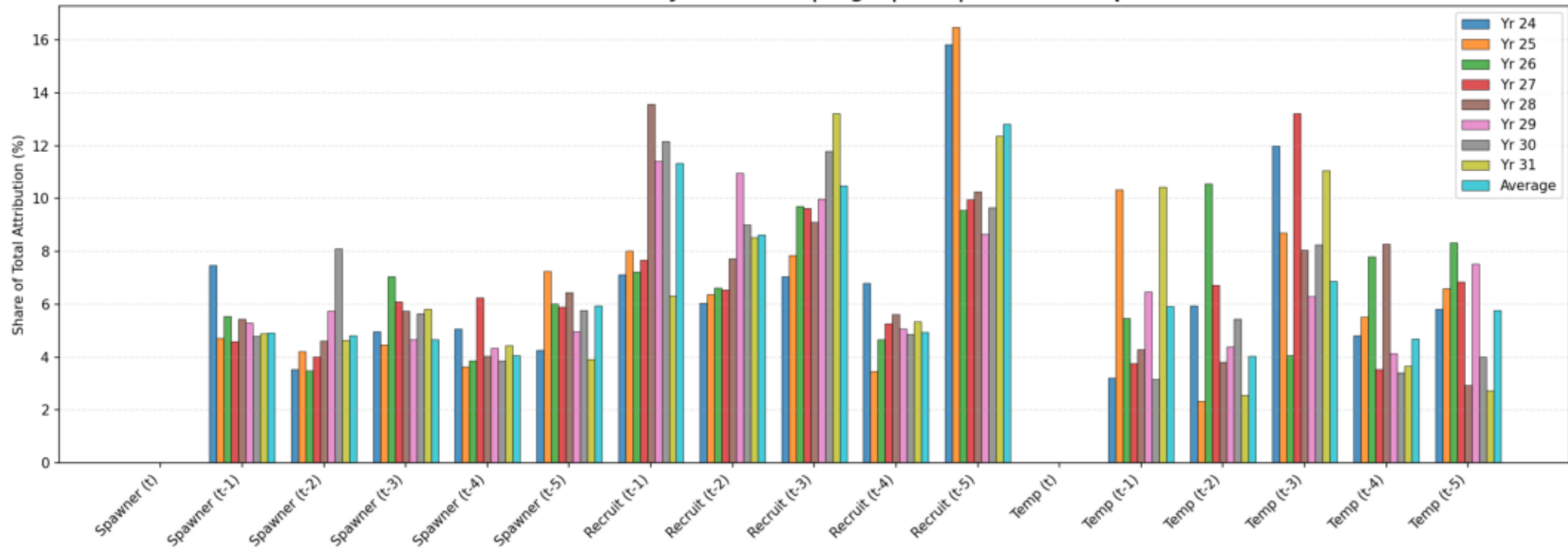
Spatial Trajectory: REAL | Lag 0 | MSE | NO Spawner(t)



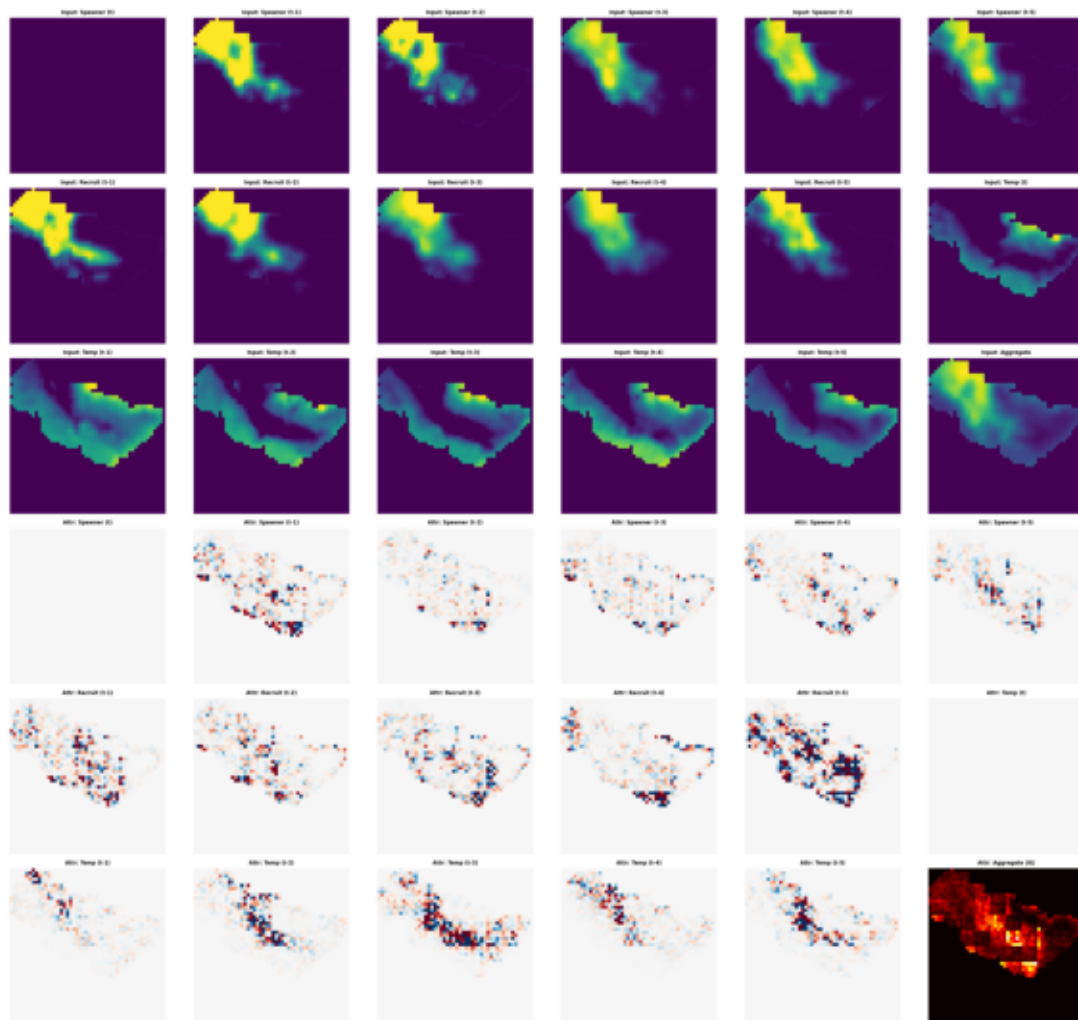
Year AVG -- REAL | Lag 0 | MSE | NO Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Channel Attribution by Year: REAL | Lag 0 | MSE | NO Current Spawner

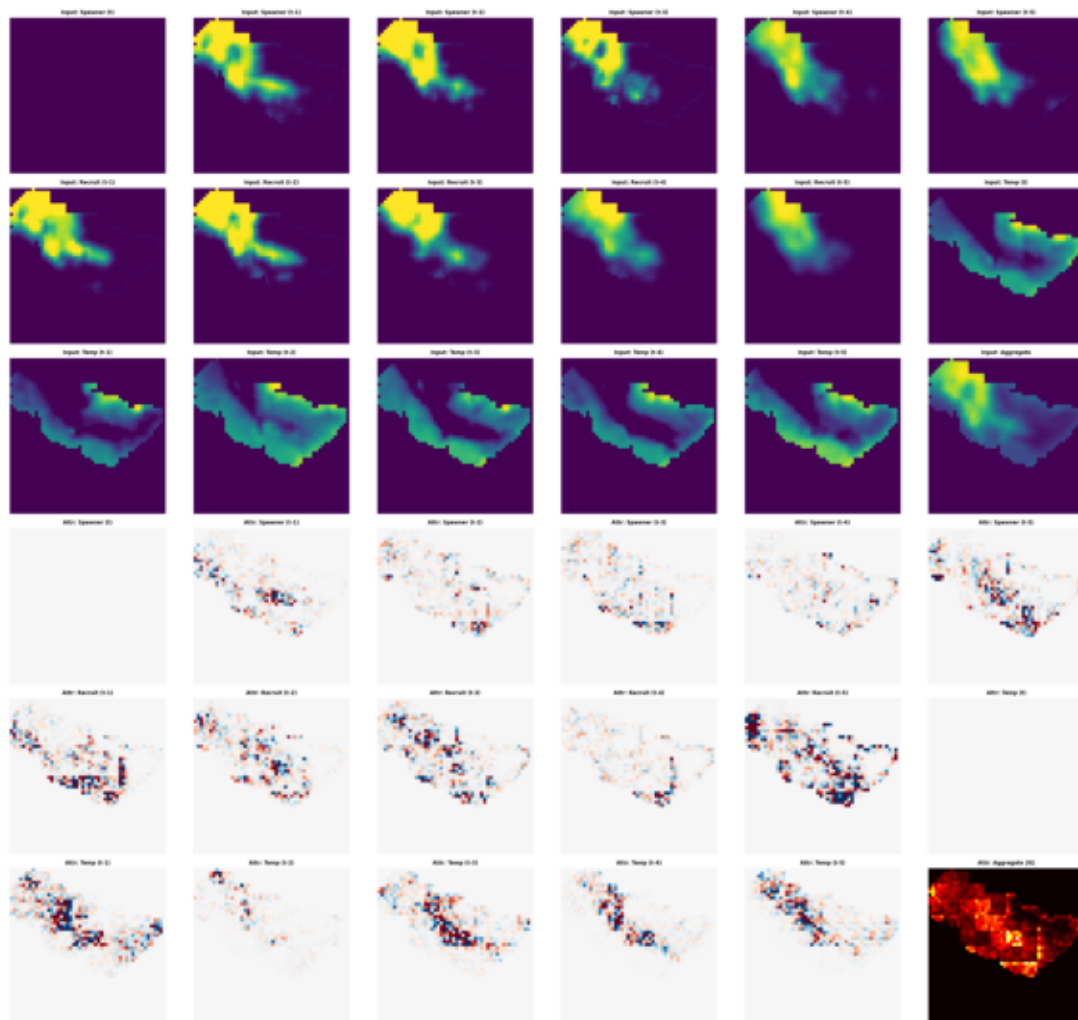


Year 24 — REAL | Lag 0 | MSE | NO Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)

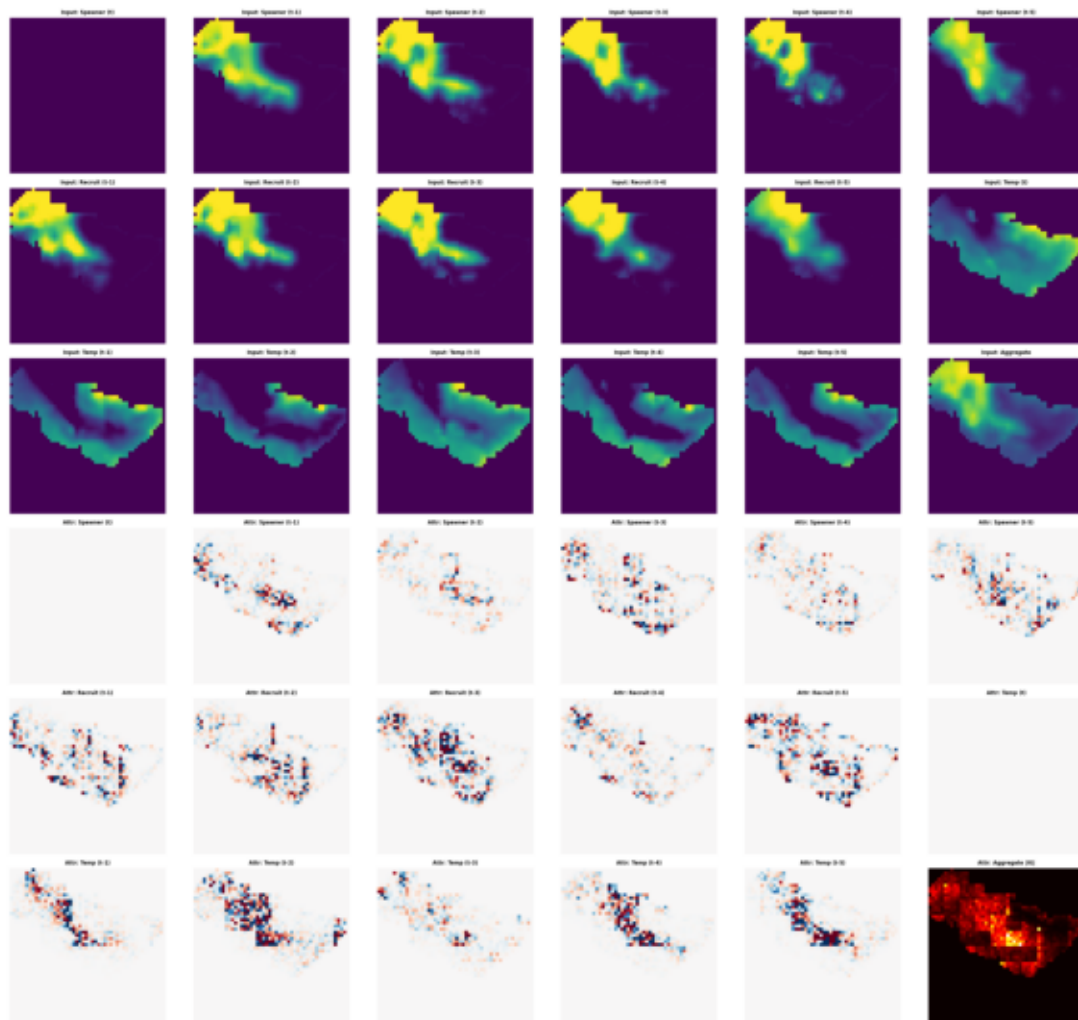




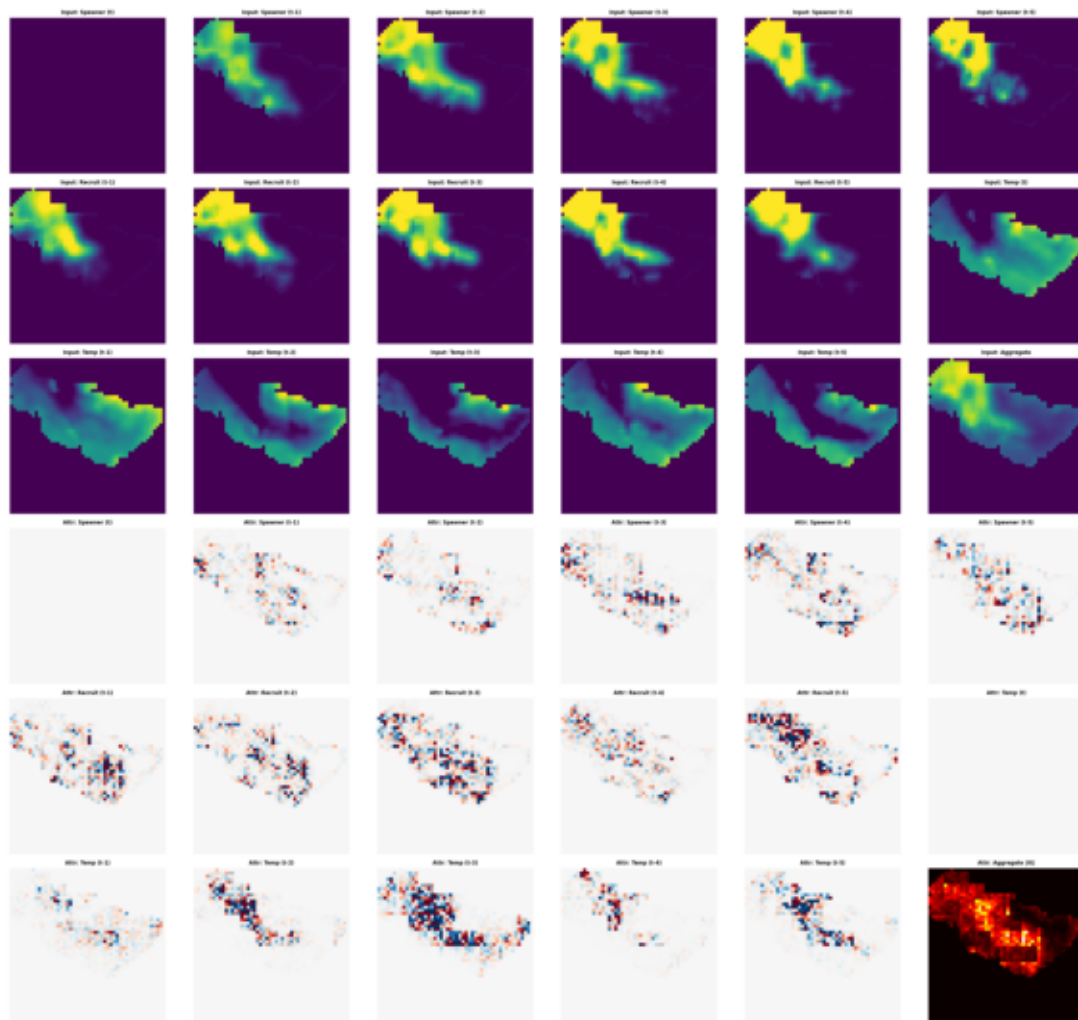
Year 25 — REAL | Lag 0 | MSE | NO Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



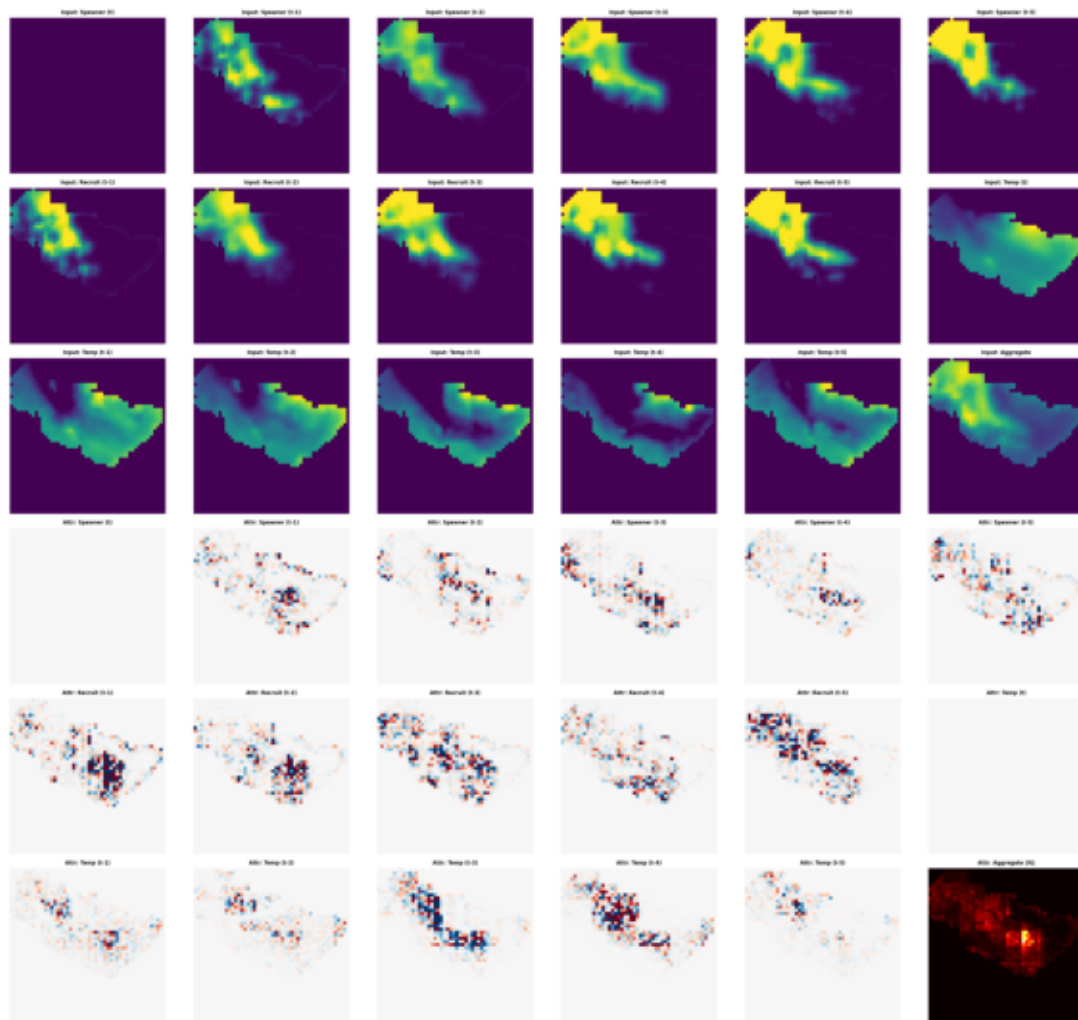
Year 26 — REAL | Lag 0 | MSE | NO Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



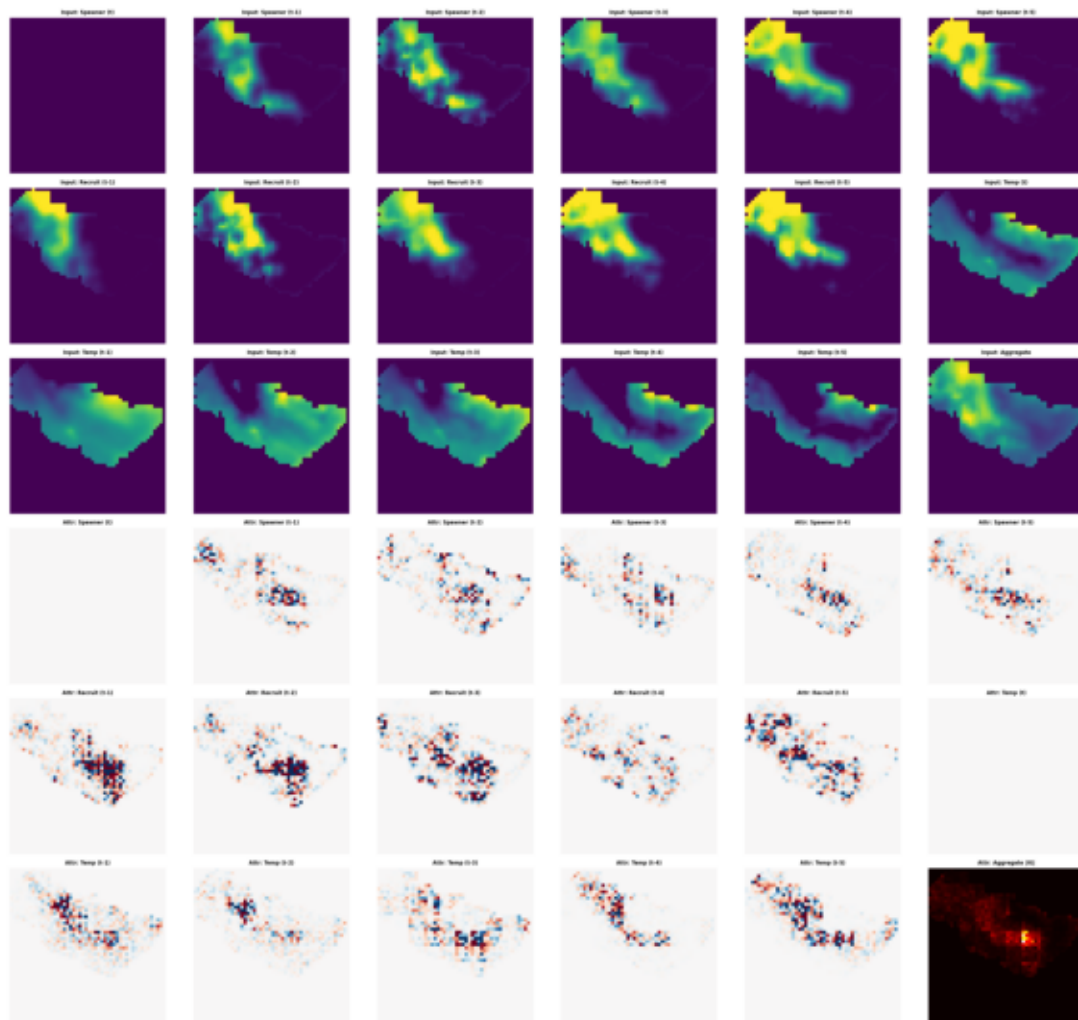
Year 27 — REAL | Lag 0 | MSE | NO Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



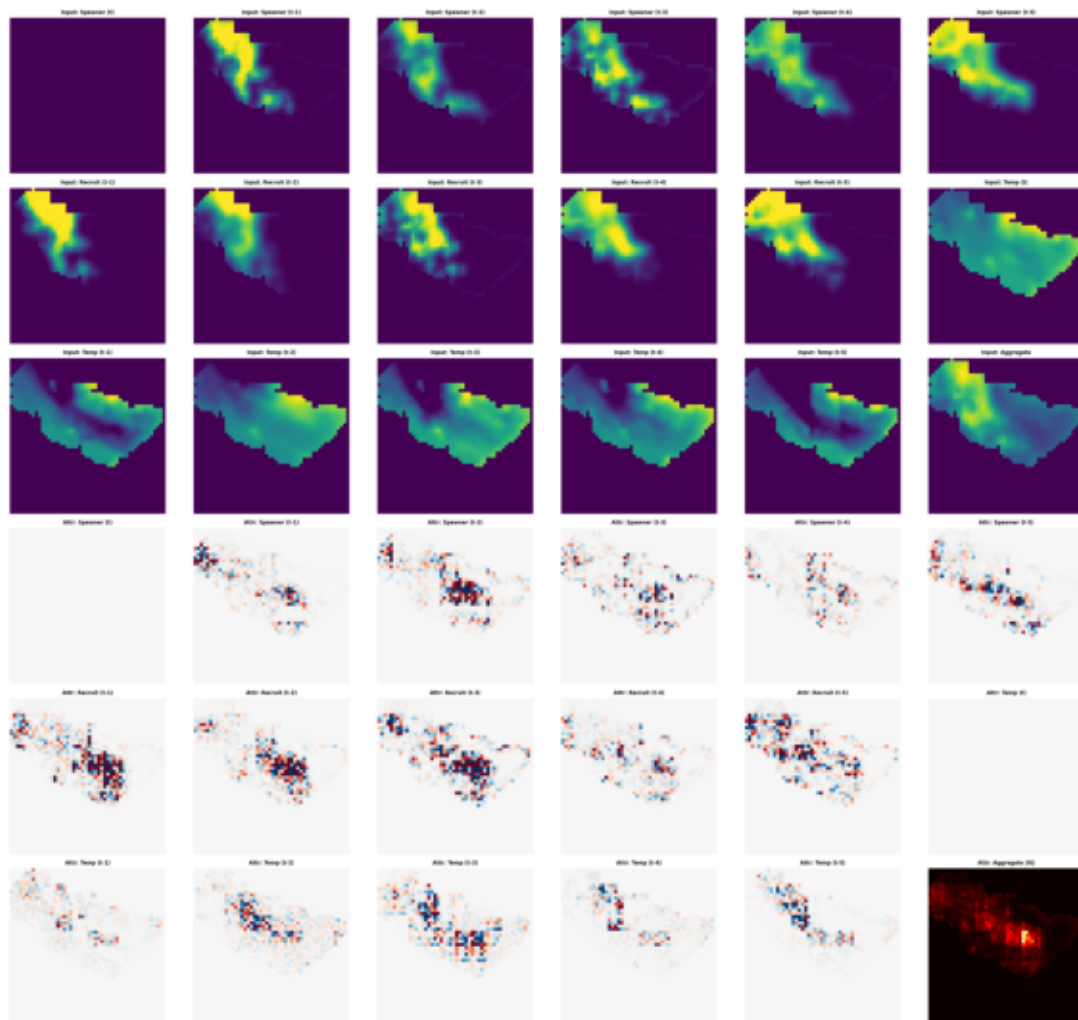
Year 28 — REAL | Lag 0 | MSE | NO Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



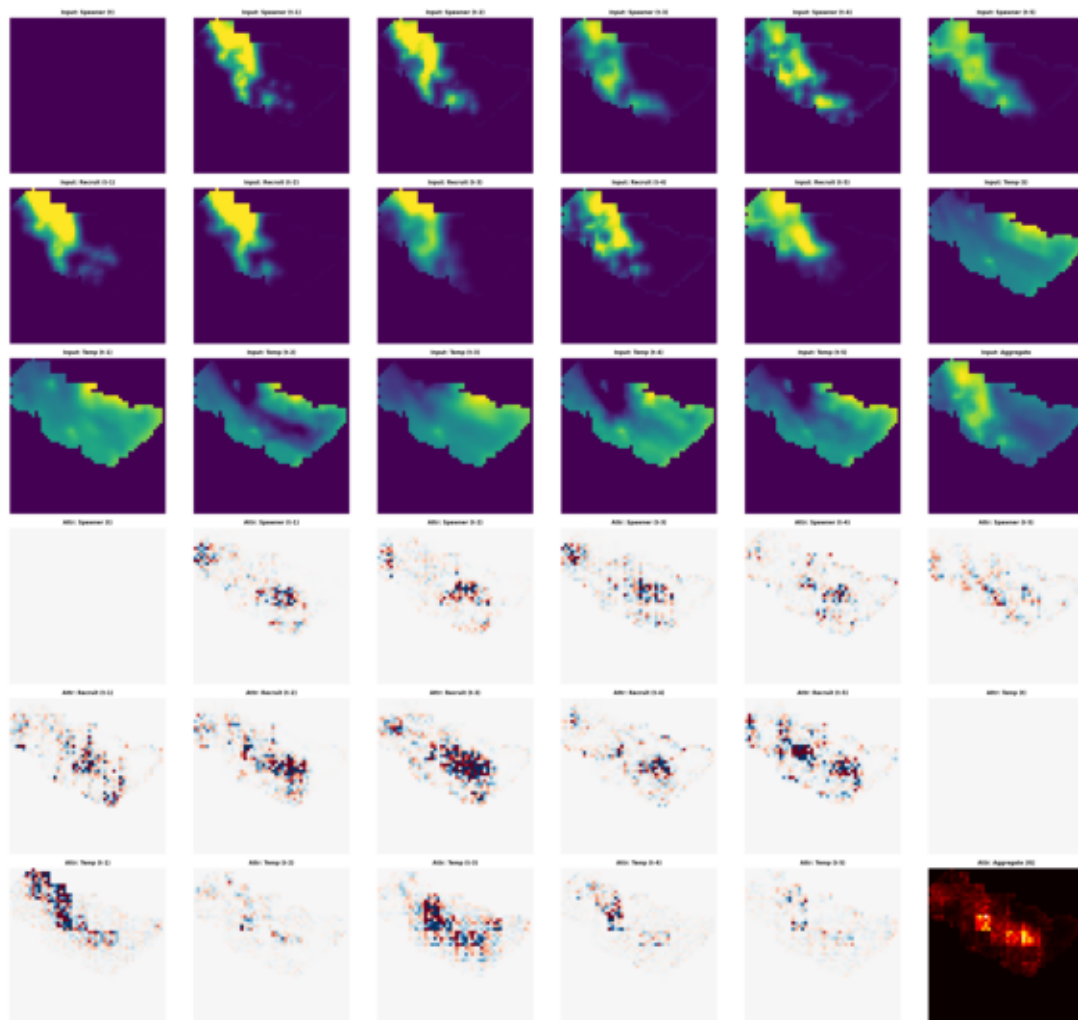
Year 29 — REAL | Lag 0 | MSE | NO Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 30 — REAL | Lag 0 | MSE | NO Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 31 — REAL | Lag 0 | MSE | NO Current Spawner  
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: REAL | Lag 0 | MSE | NO Spawner(t)

| Phase | Pred_Mean | Tgt_Mean | Pred_SD | Pred_P50 | Pred_P95 | Abundance_Capture | MAE    | Spearman |
|-------|-----------|----------|---------|----------|----------|-------------------|--------|----------|
| TRAIN | 1674.6    | 1542.8   | 6722.1  | 0.0      | 9495.4   | 1.085             | 426.5  | 0.977    |
| VAL   | 1591.7    | 1801.5   | 4756.6  | 0.0      | 12792.0  | 0.884             | 1962.9 | 0.844    |
| TEST  | 303.8     | 406.3    | 1354.3  | 0.0      | 1415.6   | 0.748             | 392.0  | 0.854    |



Detailed Metrics (R-style): REAL | Lag 0 | MSE | NO Spawner(t)

| Phase | RMSE   | RMSE_unbias | Bias   | Bias_% | Spearman | Pearson | Pct_Error | Zero_F1 | TopK_Acc | TopK_Jaccard | Quant_Err | Skill_MSE | Skill_MAE |
|-------|--------|-------------|--------|--------|----------|---------|-----------|---------|----------|--------------|-----------|-----------|-----------|
| TRAIN | 1183.5 | 1170.4      | 131.8  | 2.1%   | 0.986    | 0.982   | 3.0%      | 0.987   | 0.938    | 0.884        | 0.13      | 0.938     | 0.879     |
| VAL   | 6263.8 | 6083.6      | -209.8 | 5.4%   | 0.849    | 0.619   | 61.9%     | 0.918   | 0.594    | 0.435        | 0.91      | -2.252    | 0.034     |
| TEST  | 1813.5 | 1773.6      | -102.5 | 5.1%   | 0.863    | 0.696   | 349.7%    | 0.942   | 0.720    | 0.582        | 0.67      | -129.783  | -1.875    |